

CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

## Chapter 9: Triangles (Congruency in Triangles)

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Two angles of a triangle are  $48^\circ$  and  $67^\circ$ . Find the third angle.
2. An exterior angle of a triangle is  $110^\circ$  and one interior opposite angle is  $45^\circ$ . Find the other two angles of the triangle.
3. Which condition proves congruency:  $AB = DE$ ,  $BC = EF$ ,  $\angle B = \angle E$ ?
4. Which condition proves congruency:  $\angle A = \angle D$ ,  $\angle C = \angle F$ ,  $AC = DF$ ?
5.  $\triangle ABC \cong \triangle RQP$ . Which side equals  $AB$  and which angle equals  $\angle C$ ?
6. Can two triangles with angles  $50^\circ$ ,  $60^\circ$ ,  $70^\circ$  each be said to be congruent?
7. In triangle  $ABC$ ,  $AD$  is perpendicular to  $BC$  and  $BD = DC$ . Prove that  $AB = AC$ .
8.  $\triangle ABC \cong \triangle DEF$ ,  $BC = 2x - 3$ ,  $EF = 11$ ,  $\angle A = 55^\circ$ ,  $\angle D = (y + 15)^\circ$ . Find  $x$  and  $y$ .
9. In right triangles  $PQR$  and  $XYZ$ ,  $\angle Q = \angle Y = 90^\circ$ ,  $PR = XZ$  and  $PQ = XY$ . Name the condition and the angle equal to  $\angle R$ .
10. A point  $P$  is 5 cm from one end of a segment  $AB$  and lies on its perpendicular bisector. How far is  $P$  from the other end?

## Answer Key

---

1.  $65^\circ$
2.  $65^\circ$  and  $70^\circ$
3. S.A.S.
4. A.S.A.
5.  $AB = RQ$  and  $\angle C = \angle P$
6. Not necessarily; equal angles do not fix the size
7.  $\triangle ADB \cong \triangle ADC$  (S.A.S.:  $BD = DC$ ,  $\angle ADB = \angle ADC = 90^\circ$ , AD common), so  $AB = AC$
8.  $x = 7, y = 40$
9. R.H.S.;  $\angle Z$
10. 5 cm